

Performance Measure of Hexapole Electromagnetic Actuators:

$\mathbf{b}(\mathbf{p}) \approx \mathbf{S}(\mathbf{p})^B \hat{\mathbf{H}}_I \mathbf{I} = {}^B \mathbf{G}_I(\mathbf{p}) \mathbf{I}$; ${}^B \mathbf{G}_I(\mathbf{p})$ is the 3×6 position-dependent actuation gain matrix for flux generation.

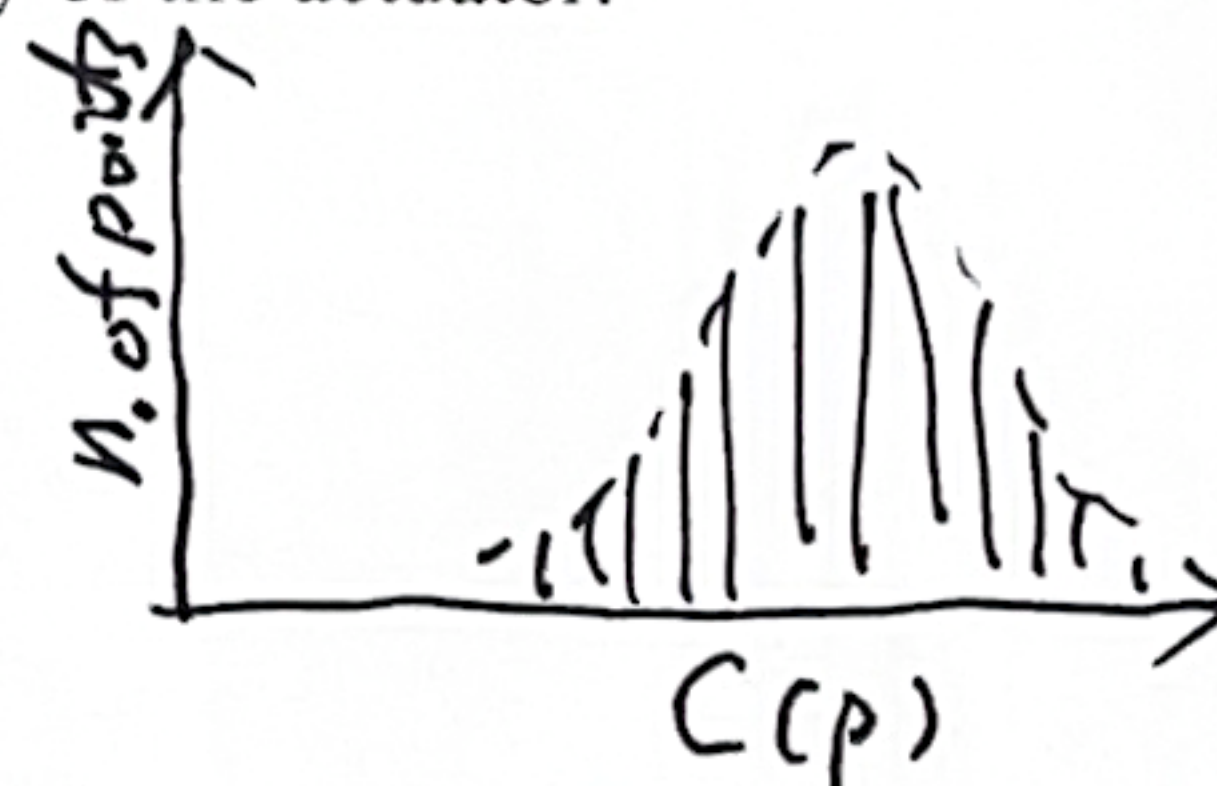
Singular Value Decomposition (SVD):

By decomposing the gain matrix via **SVD** (${}^B \mathbf{G}_I(\mathbf{p}) = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$), one obtains a set of 3 singular values: $\sigma_1 \geq \sigma_2 \geq \sigma_3$. The singular values represent the amplification factor or "gain" of the electromagnetic magnetic actuator along the three orthogonal singular vectors in the three-dimensional workspace.

Volume of the Flux Generation Ellipsoid:

In the six-input system, the input current vector constrained by a unit sphere ($\|\mathbf{I}\|_2 \leq 1$) maps to an ellipsoid in the three-dimensional flux space. A performance indicator is defined proportional to the volume of this "flux-generating ellipsoid" to measure the core capability of the actuator.

$$\mathcal{C}(\mathbf{p}) = \prod_{k=1}^3 \sigma_k(\mathbf{p})$$



Condition Number (Isotropy):

A condition number is calculated using the maximum and minimum singular values to measure how "balanced" or isotropic the actuator design is.

$$\kappa(\mathbf{p}) = \frac{\sigma_{min}}{\sigma_{max}} = \frac{\sigma_3}{\sigma_1}$$

